Introduction to Robotics

Lecture 9: Dynamics of a Single Rigid Body

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Template made by Panagiotis Papagiannopoulos

Point Mass Dynamics

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$$\boldsymbol{f} = m\boldsymbol{a}$$

where $\boldsymbol{f} \in \mathbb{R}^3$ is the sum of all forces being applied to the mass, and $\boldsymbol{a} \in \mathbb{R}^3$ is the acceleration of the point mass.

Point Mass Dynamics (2)

- Now let's assume that: 1) we have a constant gravity force being applied to the mass in the negative z-axis ($\boldsymbol{f}_g$), and 2) we can apply a force directly to the mass ($\boldsymbol{u}$).
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$$\begin{aligned} \boldsymbol{f}_g + \boldsymbol{u} &= m\boldsymbol{a} \\ \boldsymbol{u} &= m\boldsymbol{a} - \boldsymbol{f}_g \end{aligned}$$

- We can further "simplify" this equation knowing that $\boldsymbol{f}_g = \begin{bmatrix} 0 & 0 & -mg \end{bmatrix}^T$:

$$\begin{aligned} \boldsymbol{f}_g + \boldsymbol{u} &= m\boldsymbol{a} \\ \boldsymbol{u} &= m(\boldsymbol{a} + \begin{bmatrix} 0 & 0 & g \end{bmatrix}^T) \end{aligned}$$

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$$\dot{\boldsymbol{x}} = \begin{bmatrix} \dot{\boldsymbol{p}} \\ \ddot{\boldsymbol{p}} \end{bmatrix} = \begin{bmatrix} \dot{\boldsymbol{p}} \\ \boldsymbol{a} \end{bmatrix} = \begin{bmatrix} \dot{\boldsymbol{p}} \\ \frac{\boldsymbol{u}+\boldsymbol{f}_g}{m} \end{bmatrix}$$

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- Assuming a control signal $\boldsymbol{u}(t)$ and solving this differential equation, we get a $\boldsymbol{x}(t)$ function!
- In practice, solving analytically this differential equation is only possible for few systems! **And if we know the control inputs beforehand** (open-loop control)!

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- How can we solve the differential equation?
- We perform integration! Simplest form is *Euler* integration:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \dot{\boldsymbol{x}}_k dt$$

where $dt \in \mathbb{R}$ is the time step, and $\dot{\boldsymbol{x}}_k = f(\boldsymbol{x}_k, \boldsymbol{u}_k)$.

Rigid Body Dynamics

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- **Idea!** Let's re-use what we computed for the point mass!
- We can approximate a rigid body by a set of rigidly attached point masses! Total mass: $M = \sum_i m_i$.
- Each point mass is located at $\boldsymbol{r}_i = \begin{bmatrix} x_i & y_i & z_i \end{bmatrix}^T$ wrt the body frame of the rigid body.
- If $\sum_i m_i \boldsymbol{r}_i = 0$, then the origin of the body frame is called the **center of mass**. We will assume that we operate in this frame.

Rigid Body Dynamics (2)

- The rigid body is moving with a (local) twist $\mathcal{V}_b = \begin{bmatrix} \boldsymbol{\omega}_b \\ \boldsymbol{v}_b \end{bmatrix}$
- How is each point mass moving? In other words, what are the linear velocities and accelerations of the point masses?

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$$\begin{aligned}
\dot{\boldsymbol{p}}_i &= \boldsymbol{v}_b + \boldsymbol{\omega}_b \times \boldsymbol{p}_i \\
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\ddot{\boldsymbol{p}}_i &= \dot{\boldsymbol{v}}_b + \hat{\dot{\boldsymbol{\omega}}}_b \boldsymbol{p}_i + \hat{\boldsymbol{\omega}}_b \boldsymbol{v}_b + \hat{\boldsymbol{\omega}}_b^2 \boldsymbol{p}_i
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where $\boldsymbol{p}_i(t)$ are the time-varying positions of the point masses.

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  where $\boldsymbol{p}_i(t)$ are the time-varying positions of the point masses. Finally, we have (why?):

$$\ddot{\boldsymbol{p}}_i = \dot{\boldsymbol{v}}_b + \hat{\dot{\boldsymbol{\omega}}}_b \boldsymbol{r}_i + \hat{\boldsymbol{\omega}}_b \boldsymbol{v}_b + \hat{\boldsymbol{\omega}}_b^2 \boldsymbol{r}_i$$

- How is it that $\dot{\boldsymbol{p}}_i \neq 0$ when m_i is rigidly attached to the body?

Rigid Body Dynamics (3)

- We have:

$$\ddot{\boldsymbol{p}}_i = \dot{\boldsymbol{v}}_b + \hat{\dot{\omega}}_b \boldsymbol{r}_i + \hat{\omega}_b \boldsymbol{v}_b + \hat{\omega}_b^2 \boldsymbol{r}_i$$

- We can now compute the force acting on the point masses:

$$\boldsymbol{f}_i = m_i \ddot{\boldsymbol{p}}_i$$

- And in turn, this force exerts a moment/torque:

$$\begin{aligned} \boldsymbol{\tau}_i &= \boldsymbol{f}_i \times \boldsymbol{r}_i \\ &= \hat{\boldsymbol{f}}_i \boldsymbol{r}_i \end{aligned}$$

- What is the total force and moment/torque applied to the rigid body?

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- What is the total force and moment/torque applied to the rigid body?

$$\mathcal{F}_b = \begin{bmatrix} \boldsymbol{\tau}_b \\ \boldsymbol{f}_b \end{bmatrix} = \begin{bmatrix} \sum_i \boldsymbol{\tau}_i \\ \sum_i \boldsymbol{f}_i \end{bmatrix}$$

Rigid Body Dynamics (4)

- Doing calculations we can arrive at:

$$\boldsymbol{f}_b = M(\dot{\boldsymbol{v}}_b + \hat{\boldsymbol{\omega}}_b \boldsymbol{v}_b)$$
$$\boldsymbol{\tau}_b = \mathcal{I}_b \dot{\boldsymbol{\omega}}_b + \hat{\boldsymbol{\omega}}_b \mathcal{I}_b \boldsymbol{\omega}_b$$

where $\mathcal{I}_b = -\sum_i m\hat{\boldsymbol{r}}_i^2$ is the **rotational inertia matrix**.

- $\mathcal{I}_b$ is symmetric and positive definite:

$$\mathcal{I}_b = \begin{bmatrix} \sum_i m_i(y_i^2 + z_i^2) & -\sum_i m_i x_i y_i & -\sum_i m_i x_i z_i \\ -\sum_i m_i x_i y_i & \sum_i m_i(x_i^2 + z_i^2) & -\sum_i m_i y_i z_i \\ -\sum_i m_i x_i z_i & -\sum_i m_i y_i z_i & \sum_i m_i(x_i^2 + y_i^2) \end{bmatrix}$$
$$= \begin{bmatrix} \mathcal{I}_{xx} & \mathcal{I}_{xy} & \mathcal{I}_{xz} \\ \mathcal{I}_{xy} & \mathcal{I}_{yy} & \mathcal{I}_{yz} \\ \mathcal{I}_{xz} & \mathcal{I}_{yz} & \mathcal{I}_{zz} \end{bmatrix}$$

Inertia Matrices for Regular Shapes

- We now have an equation that does not depend (almost) on the point masses:

$$\mathcal{F}_b = \begin{bmatrix} \mathcal{I}_b \dot{\boldsymbol{\omega}}_b + \hat{\boldsymbol{\omega}}_b \mathcal{I}_b \boldsymbol{\omega}_b \\ M(\dot{\boldsymbol{v}}_b + \hat{\boldsymbol{\omega}}_b \boldsymbol{v}_b) \end{bmatrix}$$

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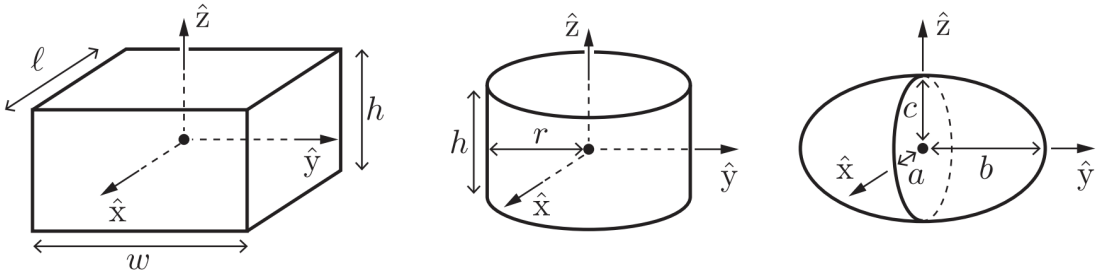
- Only $\mathcal{I}_b$ depends on the point masses (M is the total mass)
- It turns out that the rotational inertia matrix is characteristic of any rigid body
- And we can compute it for regular shapes analytically!

Inertia Matrices for Regular Shapes (2)



Source: Modern Robotics: Mechanics, Planning, and Control, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press.

- **Rectangular Parallelepiped:**

$$\mathcal{I}_{xx} = \frac{M(w^2 + h^2)}{12}, \mathcal{I}_{yy} = \frac{M(l^2 + h^2)}{12}, \mathcal{I}_{zz} = \frac{M(w^2 + l^2)}{12}$$

- **Circular Cylinder:**

$$\mathcal{I}_{xx} = \frac{M(3r^2 + h^2)}{12}, \mathcal{I}_{yy} = \frac{M(3r^2 + h^2)}{12}, \mathcal{I}_{zz} = \frac{Mr^2}{2}$$

- **Ellipsoid:**

$$\mathcal{I}_{xx} = \frac{M(b^2 + c^2)}{5}, \mathcal{I}_{yy} = \frac{M(a^2 + c^2)}{5}, \mathcal{I}_{zz} = \frac{M(a^2 + b^2)}{5}$$

More at: `https://en.wikipedia.org/wiki/List_of_moments_of_inertia`

Inertia Matrices in Different Frames

- We have $\mathcal{I}_b$ in $\{b\}$ frame
- How can we express it in frame $\{c\}$ if we know $\boldsymbol{R}_{bc}$?

Inertia Matrices in Different Frames

- We have $\mathcal{I}_b$ in $\{b\}$ frame
- How can we express it in frame $\{c\}$ if we know $\boldsymbol{R}_{bc}$?
- Kinetic energy is independent of the frame chosen:

$$\begin{aligned}\frac{1}{2}\boldsymbol{\omega}_c^T\mathcal{I}_c\boldsymbol{\omega}_c &= \frac{1}{2}\boldsymbol{\omega}_b^T\mathcal{I}_b\boldsymbol{\omega}_b \\ &= \frac{1}{2}(\boldsymbol{R}_{bc}\boldsymbol{\omega}_c)^T\mathcal{I}_b\boldsymbol{R}_{bc}\boldsymbol{\omega}_c \\ &= \frac{1}{2}\boldsymbol{\omega}_c^T\boldsymbol{R}_{bc}^T\mathcal{I}_b\boldsymbol{R}_{bc}\boldsymbol{\omega}_c\end{aligned}$$

- Thus, $\mathcal{I}_c = \boldsymbol{R}_{bc}^T\mathcal{I}_b\boldsymbol{R}_{bc}$.

Steiner's Theorem

- Many times it is convenient to represent the inertia matrix in a different frame than the center of mass
- For example, we have a composite rigid body: aka, a rigid body that is a collection of rigidly attached rigid bodies
- **Steiner's Theorem** states: *The inertia matrix $\mathcal{I}_q$ about a frame aligned with $\{b\}$, but at a point $\boldsymbol{q} = \begin{bmatrix} q_x & q_y & q_z \end{bmatrix}^T$ in $\{b\}$, is related to the inertia matrix $\mathcal{I}_b$ calculated at the center of mass by:*

$$\mathcal{I}_q = \mathcal{I}_b + M(\boldsymbol{q}^T\boldsymbol{q}\boldsymbol{I} - \boldsymbol{q}\boldsymbol{q}^T)$$

where $\boldsymbol{I}$ is the identity matrix and M is the mass of the body.

Twist-Wrench Formulation

- We can write the dynamics in the following form:

$$\begin{bmatrix} \boldsymbol{\tau}_b \\ \boldsymbol{f}_b \end{bmatrix} = \begin{bmatrix} \mathcal{I}_b & 0 \\ 0 & M\boldsymbol{I} \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{\omega}}_b \\ \dot{\boldsymbol{v}}_b \end{bmatrix} + \begin{bmatrix} \hat{\boldsymbol{\omega}}_b & 0 \\ 0 & \hat{\boldsymbol{\omega}}_b \end{bmatrix} \begin{bmatrix} \mathcal{I}_b & 0 \\ 0 & M\boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{\omega}_b \\ \boldsymbol{v}_b \end{bmatrix}$$

$$= \begin{bmatrix} \mathcal{I}_b & 0 \\ 0 & M\boldsymbol{I} \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{\omega}}_b \\ \dot{\boldsymbol{v}}_b \end{bmatrix} - \begin{bmatrix} \hat{\boldsymbol{\omega}}_b & 0 \\ \hat{\boldsymbol{v}}_b & \hat{\boldsymbol{\omega}}_b \end{bmatrix}^T \begin{bmatrix} \mathcal{I}_b & 0 \\ 0 & M\boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{\omega}_b \\ \boldsymbol{v}_b \end{bmatrix}$$

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- What does this give us?

Twist-Wrench Formulation (2)

Let's do some notations:

- **Spatial Inertia Matrix:** $\mathcal{G}_b = \begin{bmatrix} \mathcal{I}_b & 0 \\ 0 & M\boldsymbol{I} \end{bmatrix} \in \mathbb{R}^{6\times 6}$
- **Spatial Momentum:** $\mathcal{P}_b = \mathcal{G}_b \mathcal{V}_b \in \mathbb{R}^6$
- **Adjoint operator:** $[ad_{\mathcal{V}}] = \begin{bmatrix} \hat{\boldsymbol{\omega}} & 0 \\ \hat{\boldsymbol{v}} & \hat{\boldsymbol{\omega}} \end{bmatrix}$, where $\mathcal{V} = \begin{bmatrix} \boldsymbol{\omega} \\ \boldsymbol{v} \end{bmatrix}$

Now we can write the dynamics equation as:

$$\mathcal{F}_b = \mathcal{G}_b \dot{\mathcal{V}}_b - [ad_{\mathcal{V}_b}]^T \mathcal{G}_b \mathcal{V}_b$$

Forward and Inverse Dynamics

- **Inverse Dynamics:** Given a desired acceleration $\dot{\mathcal{V}}_b$, we can compute the wrench to achieve it by

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  - We answer the question: *"What is the wrench required to exert in order to achieve a desired acceleration?"*
- **Forward Dynamics:** Given a realized wrench $\mathcal{F}_b$, we can compute the acceleration generated by it

$$\dot{\mathcal{V}}_b = \mathcal{G}_b^{-1}\left(\mathcal{F}_b + [ad_{\mathcal{V}_b}]^T \mathcal{G}_b \mathcal{V}_b\right)$$

  - We can use this for forward simulation or estimation
  - We answer the question: *"What is the acceleration that is generated by the application of a specific wrench?"*

Dynamics in different frames

We have been deriving all equations in center of mass of frame $\{b\}$. We can express the dynamics in other frames. Let's call this frame $\{a\}$. The main "conversion" that we need is to transform the spatial inertia matrix from one frame to the other. We can do the same trick with the kinetic energy:

$$\begin{aligned}\frac{1}{2}\mathcal{V}_a^T\mathcal{G}_a\mathcal{V}_a &= \frac{1}{2}\mathcal{V}_b^T\mathcal{G}_b\mathcal{V}_b \\ &= \frac{1}{2}([Ad_{\boldsymbol{T}_{ba}}]\mathcal{V}_a)^T\mathcal{G}_b[Ad_{\boldsymbol{T}_{ba}}]\mathcal{V}_a \\ &= \frac{1}{2}\mathcal{V}_a^T[Ad_{\boldsymbol{T}_{ba}}]^T\mathcal{G}_b[Ad_{\boldsymbol{T}_{ba}}]\mathcal{V}_a\end{aligned}$$

Thus, we have $\mathcal{G}_a = [Ad_{\boldsymbol{T}_{ba}}]^T\mathcal{G}_b[Ad_{\boldsymbol{T}_{ba}}]$, and:

$$\mathcal{F}_a = \mathcal{G}_a\dot{\mathcal{V}}_a - [ad_{\mathcal{V}_a}]^T\mathcal{G}_a\mathcal{V}_a$$

Bibliography

Chapter 8 from **Modern Robotics: Mechanics, Planning, and Control**, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press. ebook

Thank you

- **Any Questions?**
- **Office Hours**:
  - **Tue-Thu** *(10:00-12:00)*
  - 24/7 by email (`costashatz@upatras.gr`, subject: *ECE_IR_AM*)
- **Material and Announcements**

Laboratory of Automation & Robotics